THORACIC WALL, INTERCOSTAL SPACES AND INTERCOSTAL MUSCLES

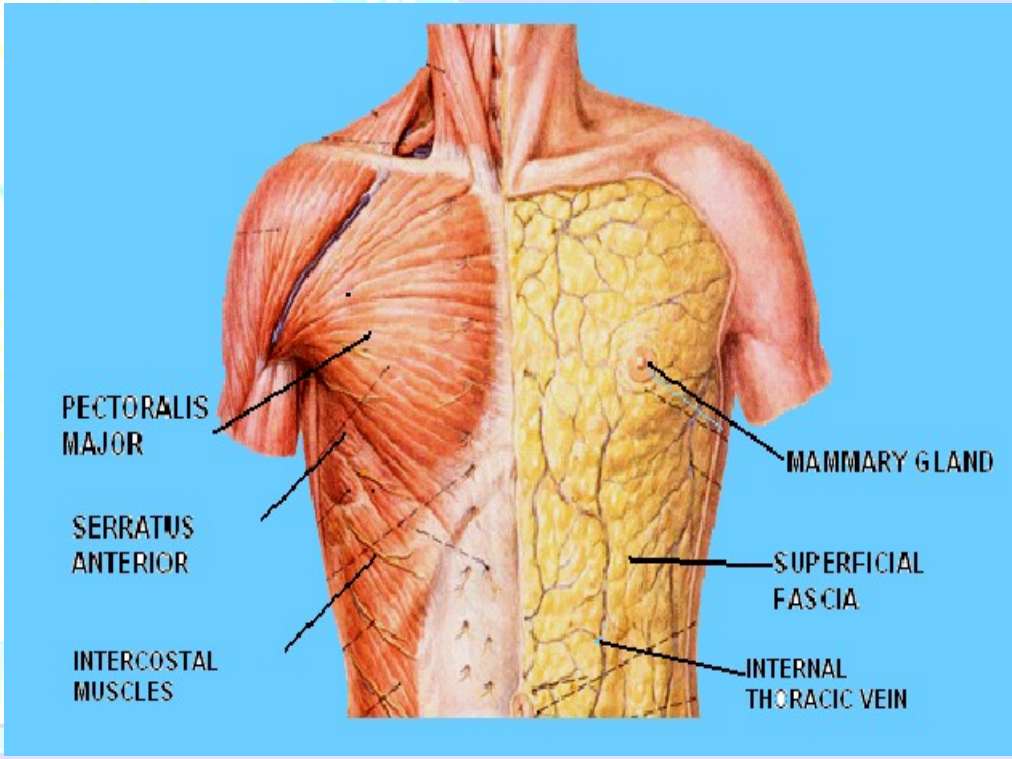
LEARNING OBJECTIVES

At the end of the lecture the student should be able to know about:

- Intercostal muscles
- Content of intercostal spaces
- Origin of intercostal arteries
- Origin, course and distribution of intercostal nerves
- Branches and course of internal thoracic artery
- Clinical correlation of thoracic wall

THORACIC WALL

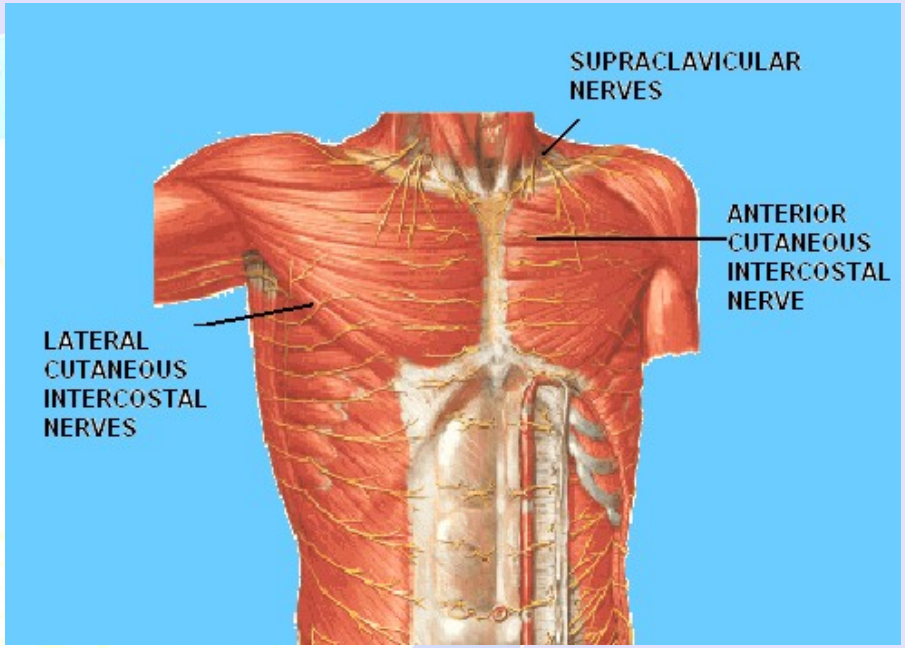
- Covered by the following structures from outside to inside
- Skin
- Superficial fascia
  - Thoracoepigastric v.
  - Supraclavicular n.
  - Anterior and lateral cutaneous branches of intercostal n.
  - Mammary gland
- Muscles
- Deep fascia– endothoracic fascia
- Parietal pleura



CUTANEOUS INNERVATION OF CHEST WALL

Anteriorly

- Above sternal angle; T4; from ***Supraclavicular nerves***
- Below sternal angle, ***anterior and lateral cutaneous branches of intercostal nerve***



Posteriorly

- Dorsal rami of spinal nerves

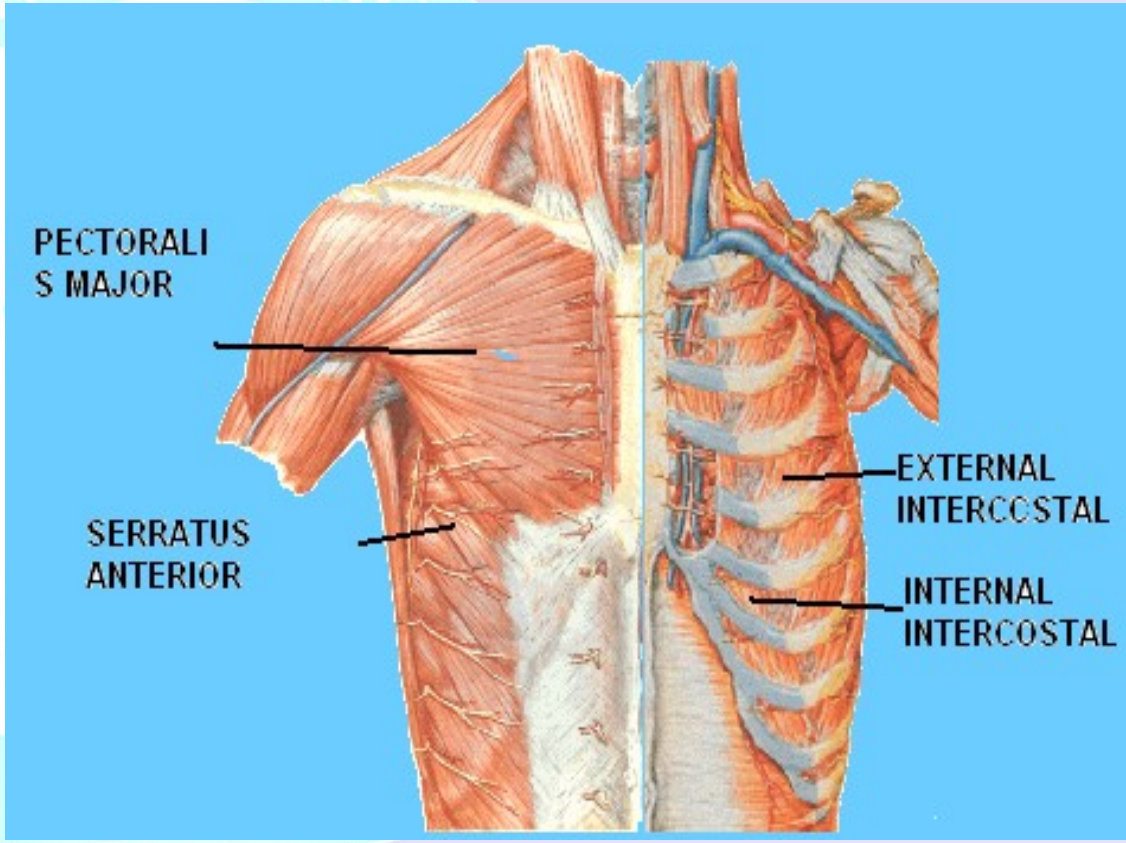
THE MUSCLES OF THORAX

Extrinsic muscles

- Pectoralis major
- Pectoralis minor
- Serratus anterior

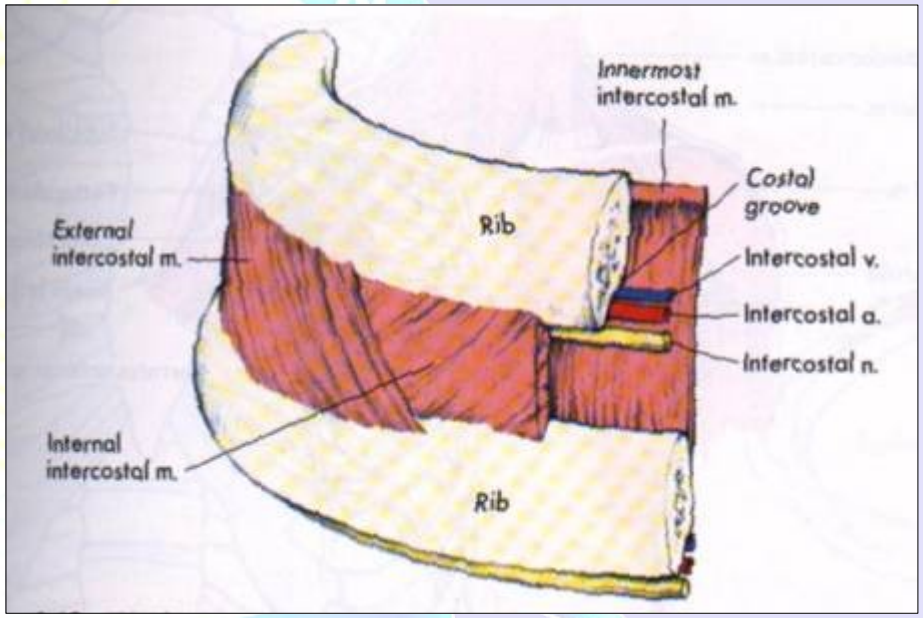
Intrinsic muscles

- Intercostales externi
- Intercostales interni
- Intercostales intimi
- Transverses thoracis



INTERCOSTAL SPACES

- Spaces between ribs
- 11 in number
- Contain three layers of muscles of respiration
  - External intercostal
  - Internal intercostal
  - Innermost inter costal
- Last one lined by endothoracic fascia and then parietal pleura
- Each intercostal space has its own neurovascular bundle



EXTERNAL INTERCOSTAL MUSCLES

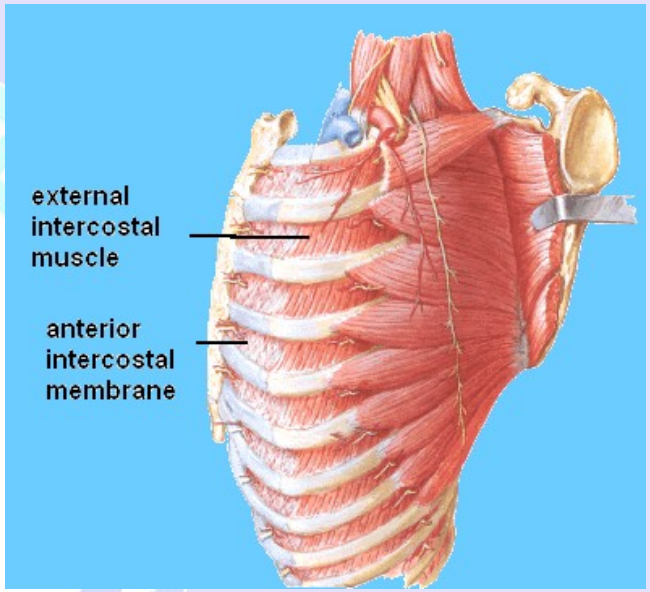
- Most superficial of intercostals muscle

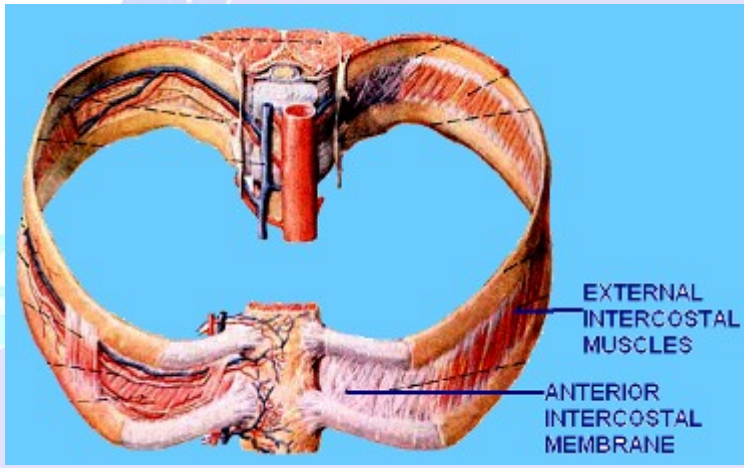
Origin

- Inferior border of the rib above
- **Insertion**
- Superior border of the rib below

Course

- Fibers run downward and forward
- Extends from the costal tubercle to the costochondral junction where it is replaced by **anterior intercostal membrane**
- **Action**: elevate ribs adding in forced inspiration





INTERNAL INTERCOSTAL MUSCLES

- Forms intermediate layer

Origin

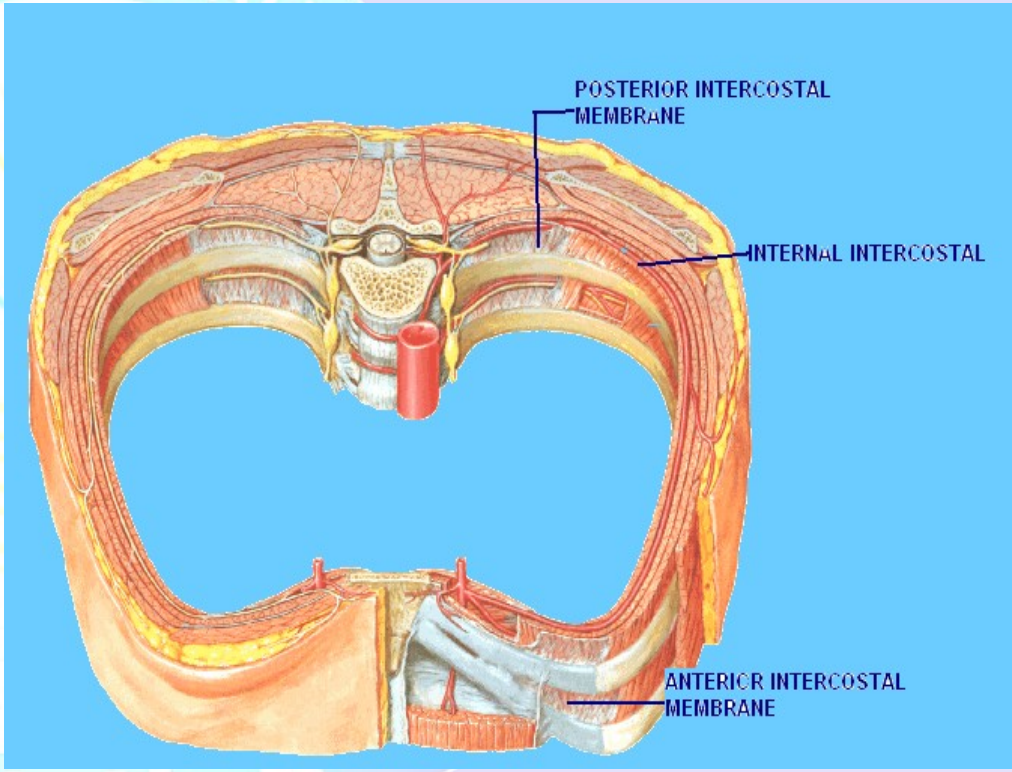
- Subcostal groove of the rib above

Insertion

- Upper border of rib below

Course

- Fibers runs downward and backward from sternum to angle of the rib
- At angle of the rib it forms ***posterior intercostal membrane***
- **Action**: depress ribs for forced expiration

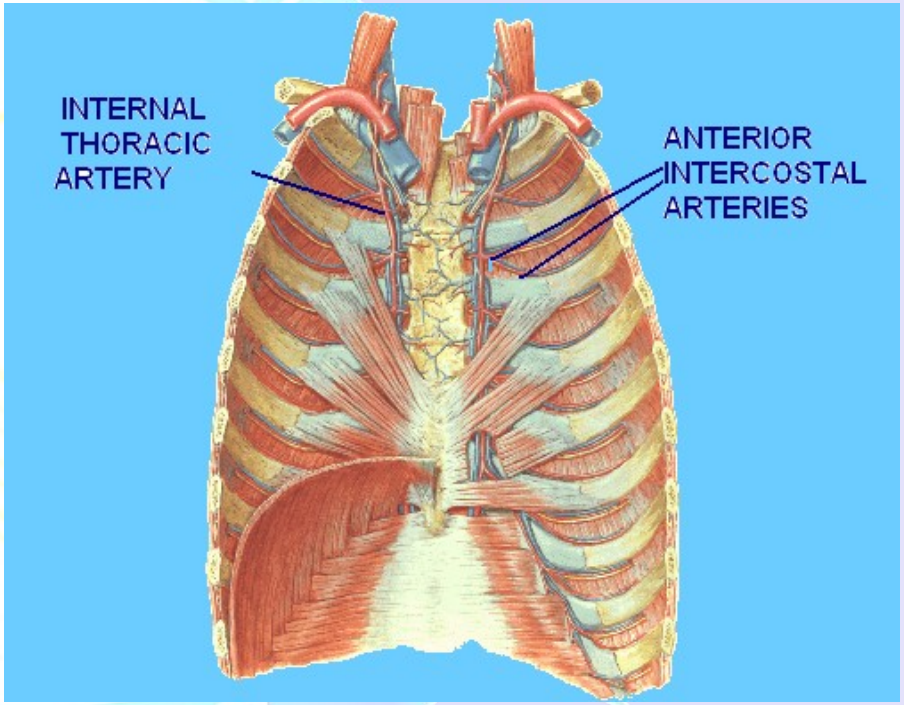


INNERMOST INTERCOSTAL

- Deepest layer
- Crosses more than one intercostal space
- Related internally to endothoracic fascia and parietal pleura
- Externally related to intercostal vessels and nerves
- Variable muscle
- Uncertain function

INTERCOSTAL ARTERIES

- Two anterior intercostal and one single posterior intercostal artery in each space



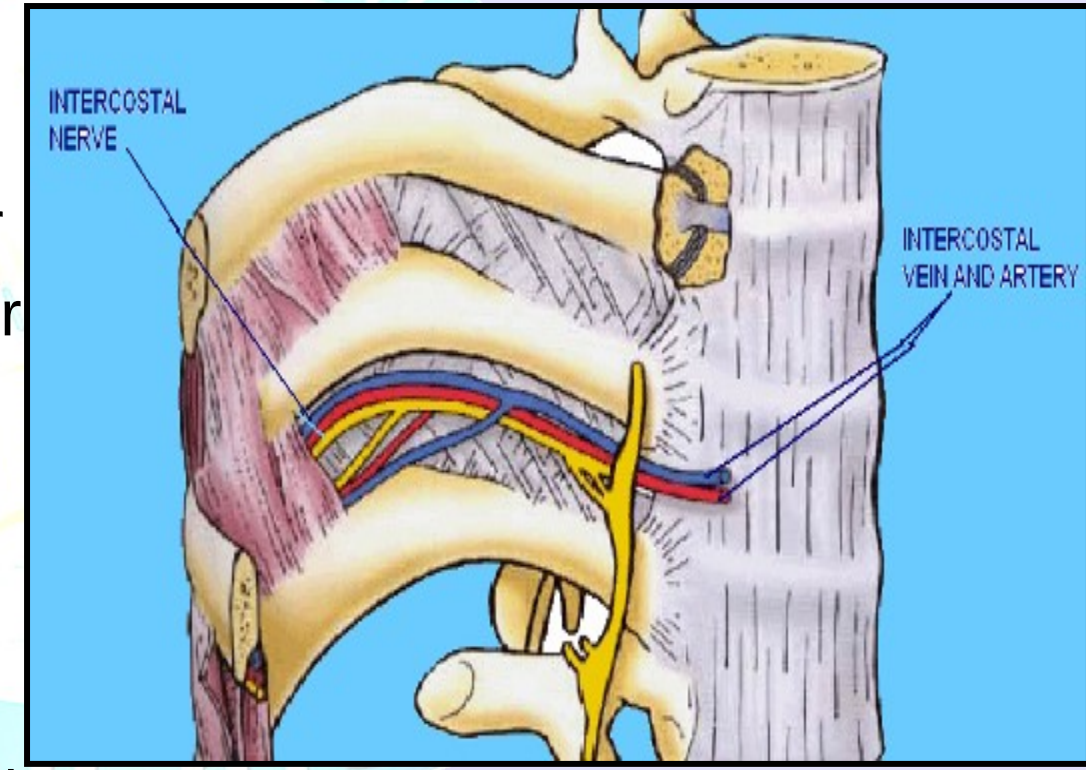
Anterior intercostal arteries

- Of first six spaces from ***internal thoracic artery***

- Of last six are the branches of ***musculophrenic artery***

NEUROVASCULAR BUNDLE OF INTERCOSTAL MUSCLES

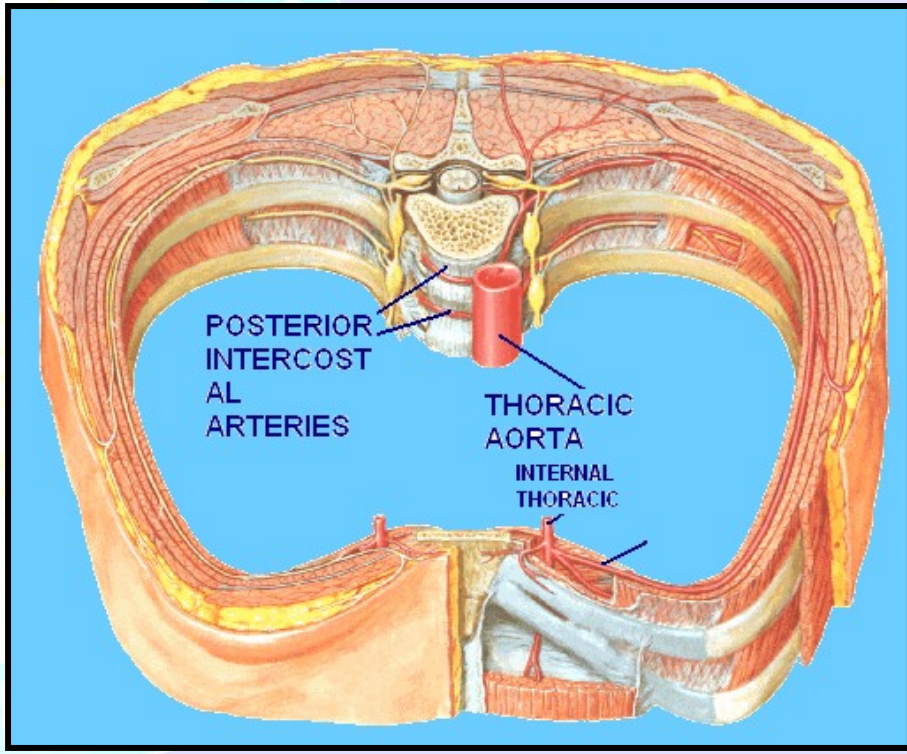
- Inter costal vessels and nerves
- Run between internal and innermost intercostal layer
- Arranged in following order from above downwards
  - **Intercostal vein**
  - **Intercostal artery**
  - **Intercostal nerve**
- Runs in subcostal groove
- Runs near the upper margin of concerned intercostal space



INTERCOSTAL ARTERIES

POSTERIOR INTERCOSTAL ARTERIES

- subclavian artery→ costocervical trunk→ **superior intercostal artery**→posterior intercostal arteries of first 2 spaces
- Those of last 9 spaces from **descending thoracic aorta**



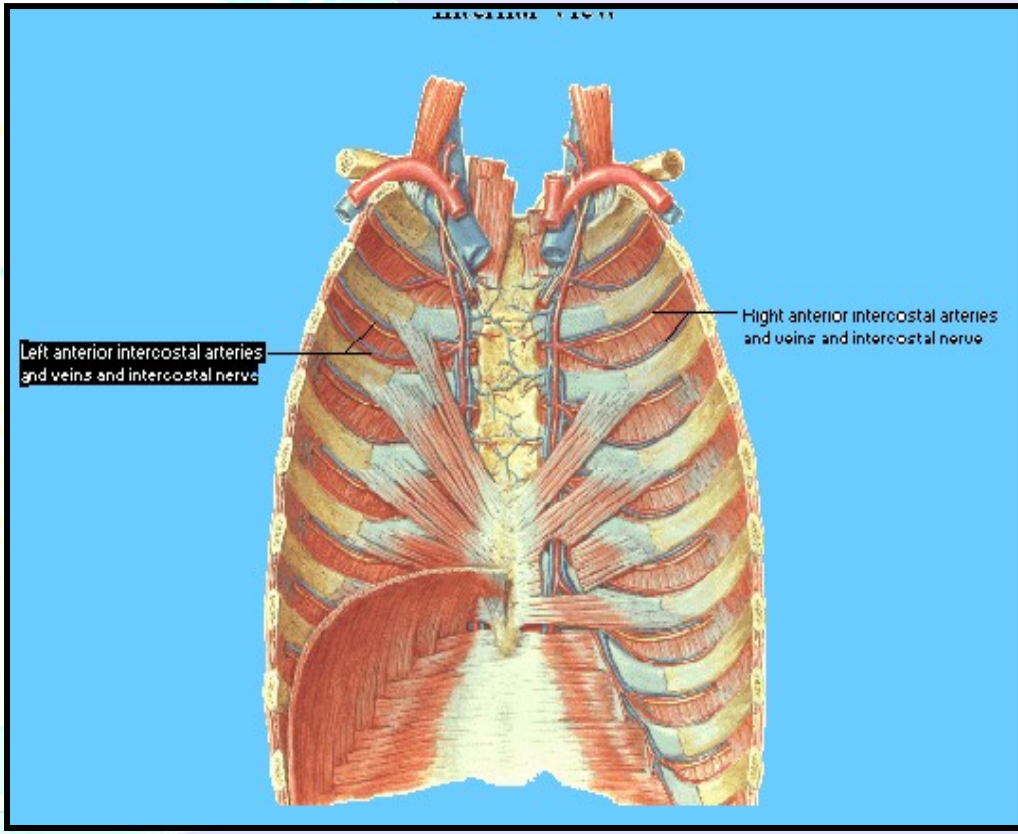
INTERCOSTAL VEINS

Posterior intercostal veins

- Drain into azygos or hemiazygos veins

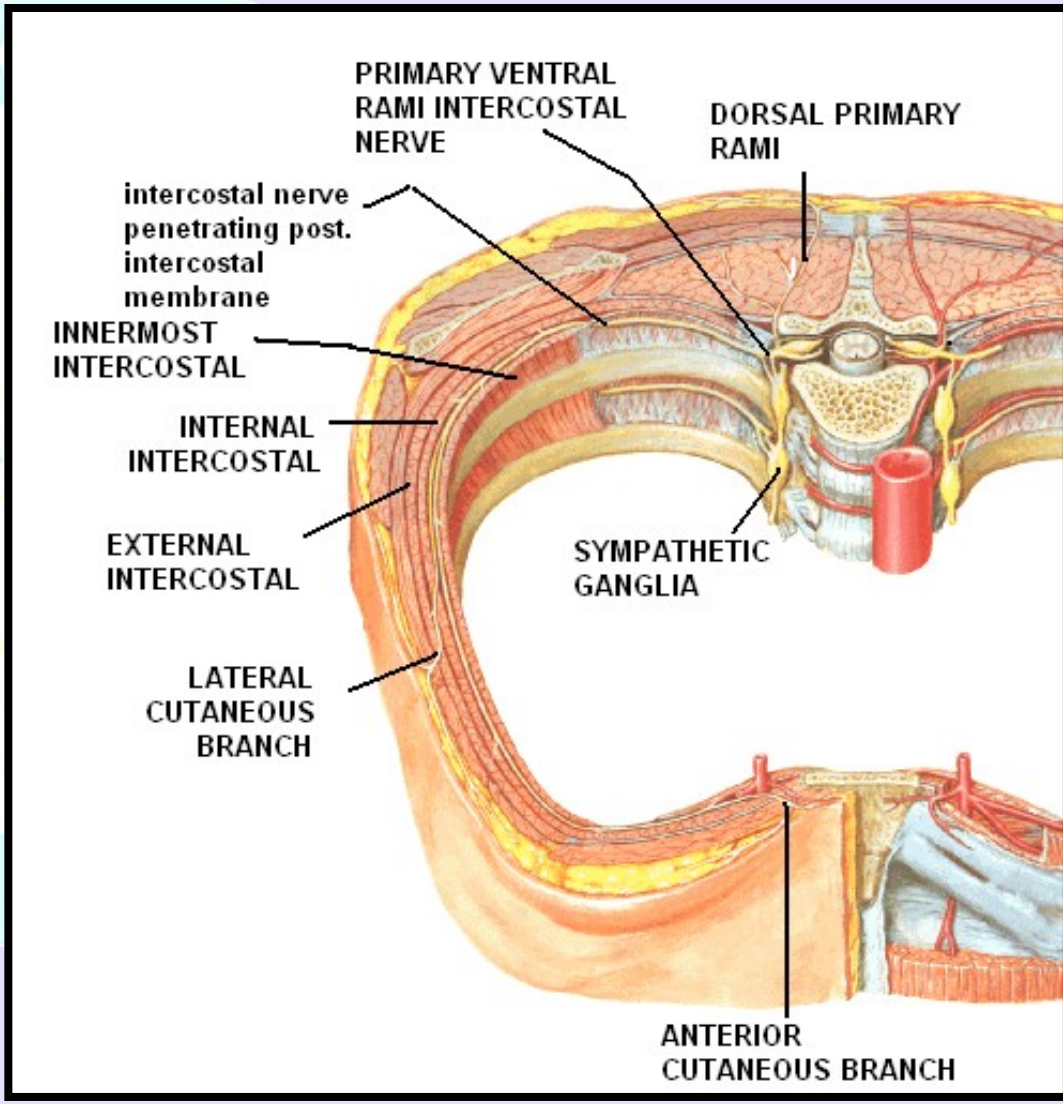
Anterior intercostal veins

- Drain into internal thoracic and musculophrenic vein



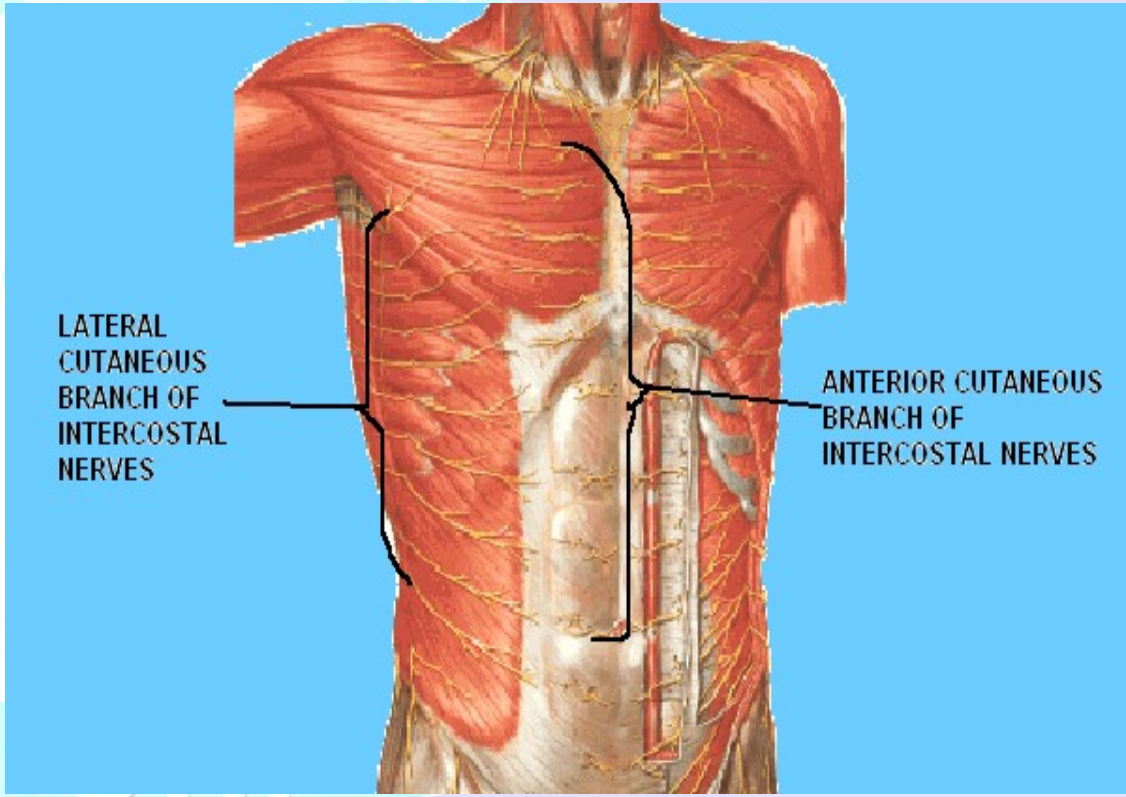
INTERCOSTAL NERVES

- Anterior primary rami of first 11 thoracic spinal nerves
- Enter intercostal space between parietal pleura and posterior intercostal membrane
- Runs at the lower border of rib between internal intercostal and inner most intercostal



INTERCOSTAL NERVES

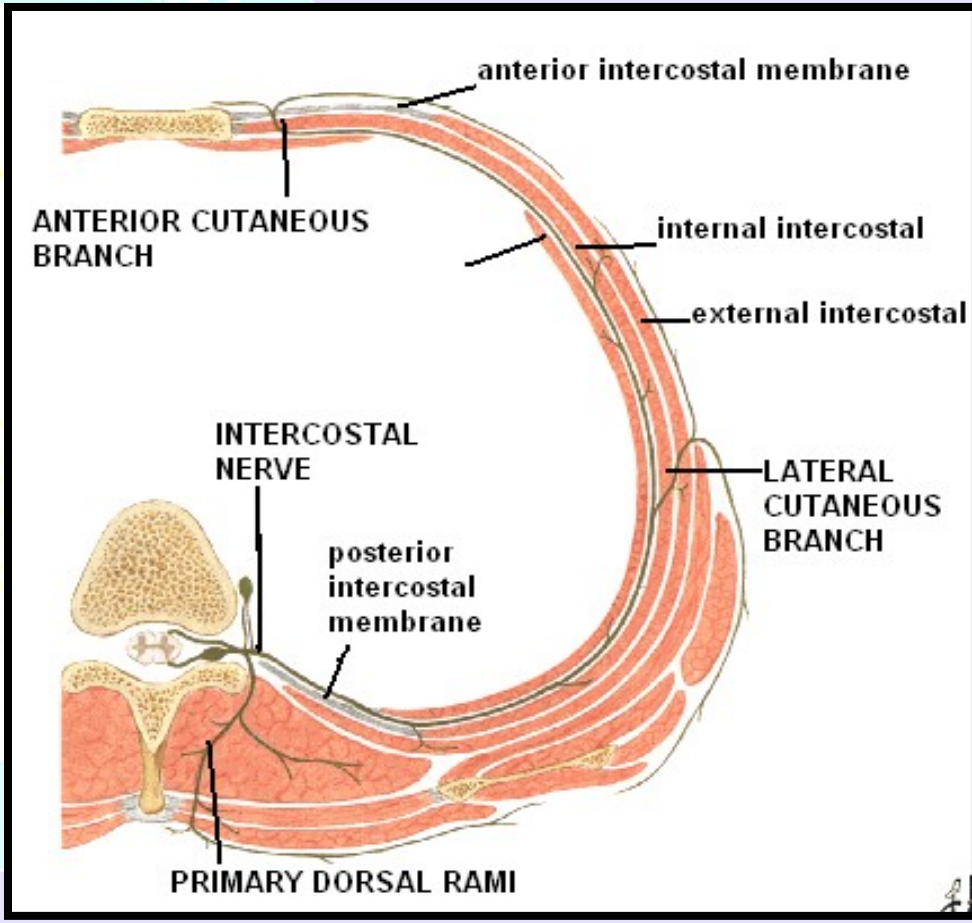
- 1st to 6th remain in their space
- 7th to 9th pass deep to costal cartilage →enter abdominal wall
- 10th and 11th enter directly abdominal wall



INTERCOSTAL NERVES

Branches

- Collateral branch→runs in the same space at upper border of rib below
- Lateral cutaneous branch
- Anterior cutaneous branch
- Muscular branches
- Pleural sensory branches
- Peritoneal sensory branches



INTERCOSTAL NERVES

DISTRIBUTION

1st to 6th supply:

- Skin and parietal pleura of intercostal spaces
- Intercostal muscles of these spaces

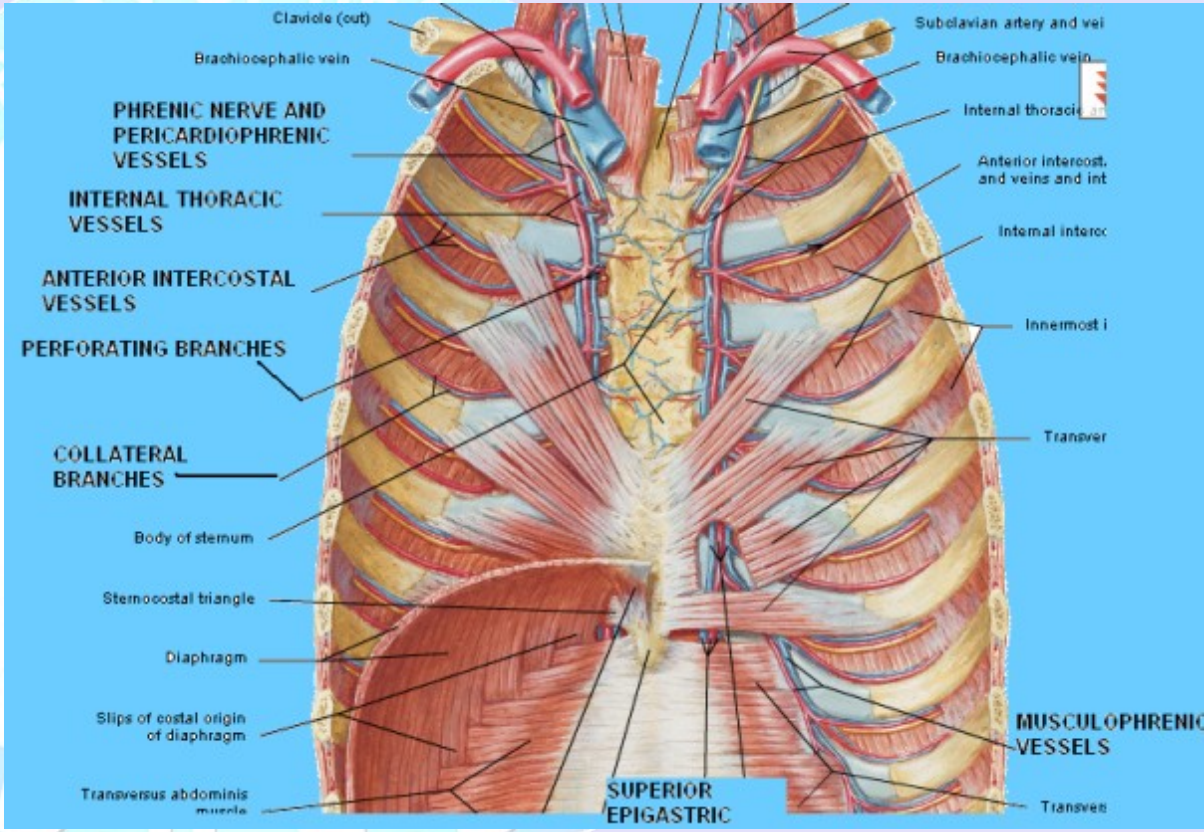
7th to 11th supply:

- Skin and parietal peritoneum of outer inner surface of abdominal wall
- Anterior abdominal wall muscles

INTERNAL THORACIC ARTERY

Branches:

- 2 Anterior intercostal arteries for each of the upper six spaces
  - Perforating arteries
  - Pericardiophrenic artery
  - Mediastinal arteries
  - Superior epigastric artery
  - Musculophrenic artery



INTERNAL THORACIC VEIN

- Accompanies internal thoracic artery
- Drains into brachiocephalic vein on each side

INTERNAL THORACIC ARTERY

- Branch of 1st part of subclavian artery in neck
- Descends vertically on on the pleura behind costal cartilage
- Ends in 6th intercostal space by dividing into ***superior epigastric and musculophrenic artery***

CLINICAL CORRELATES

Referred pain

- Disease in thorax
- Pain feel along costal margin to anterior abdominal wall
- 7th to 11th intercostal nerve supplies anterior abdominal wall

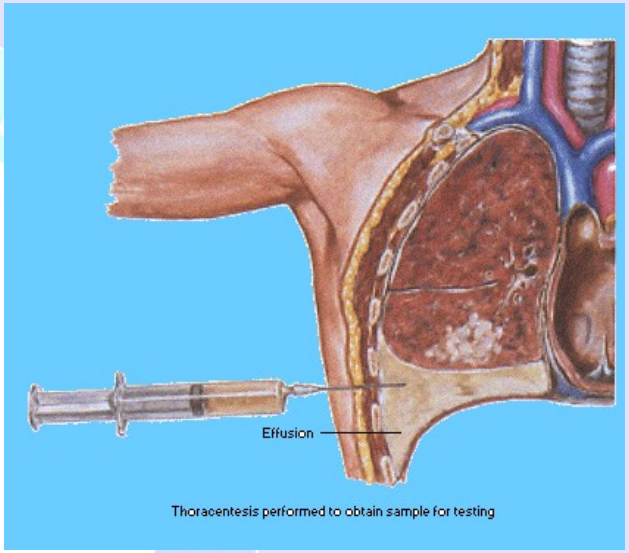
CLINICAL CORRELATES

Herpes zoster

- Latent varicella zoster infection
- Inflammation and degeneration of sensory neuron
- Formation of vesicle
- In thorax, band of pain in intercostal dermatome followed by vesicle

NEEDLE THORACOSTOMY

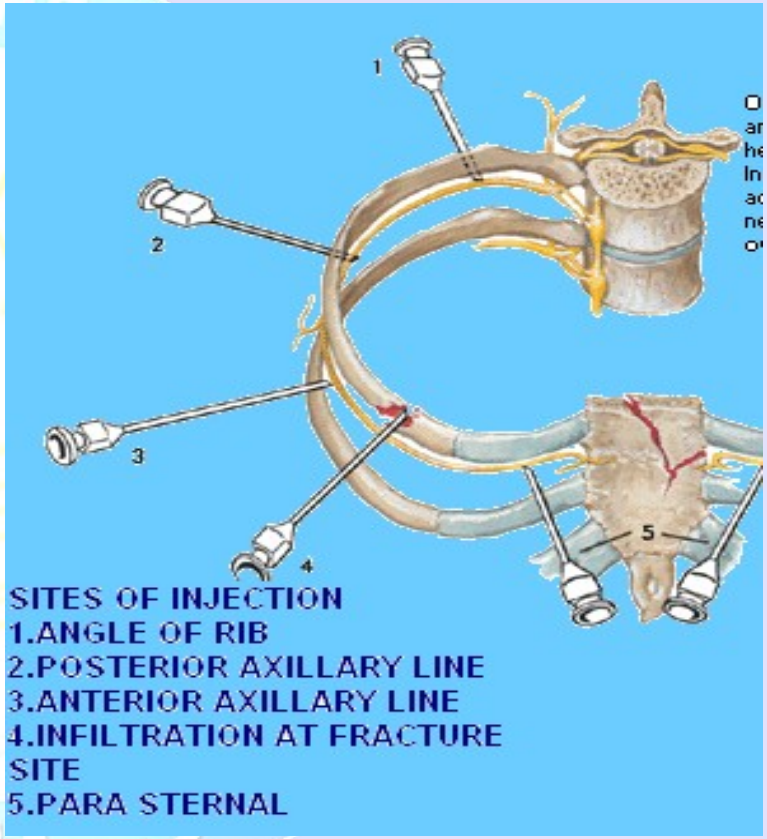
- Insertion of needle in the pleural cavity
  - Drainage of air, pus, blood
  - Drawing a sample
- Anterior or lateral approach
- Needle passes through
  - **Skin, superficial fascia**
  - **Serratus anterior**
  - **External intercostal muscles**
  - **Internal intercostal muscles**
  - **Inner most inter costal**
  - **Endothoracic fascia**
  - **Parietal pleura**
- Needle should be kept close to upper border of rib below to avoid injury to VAN



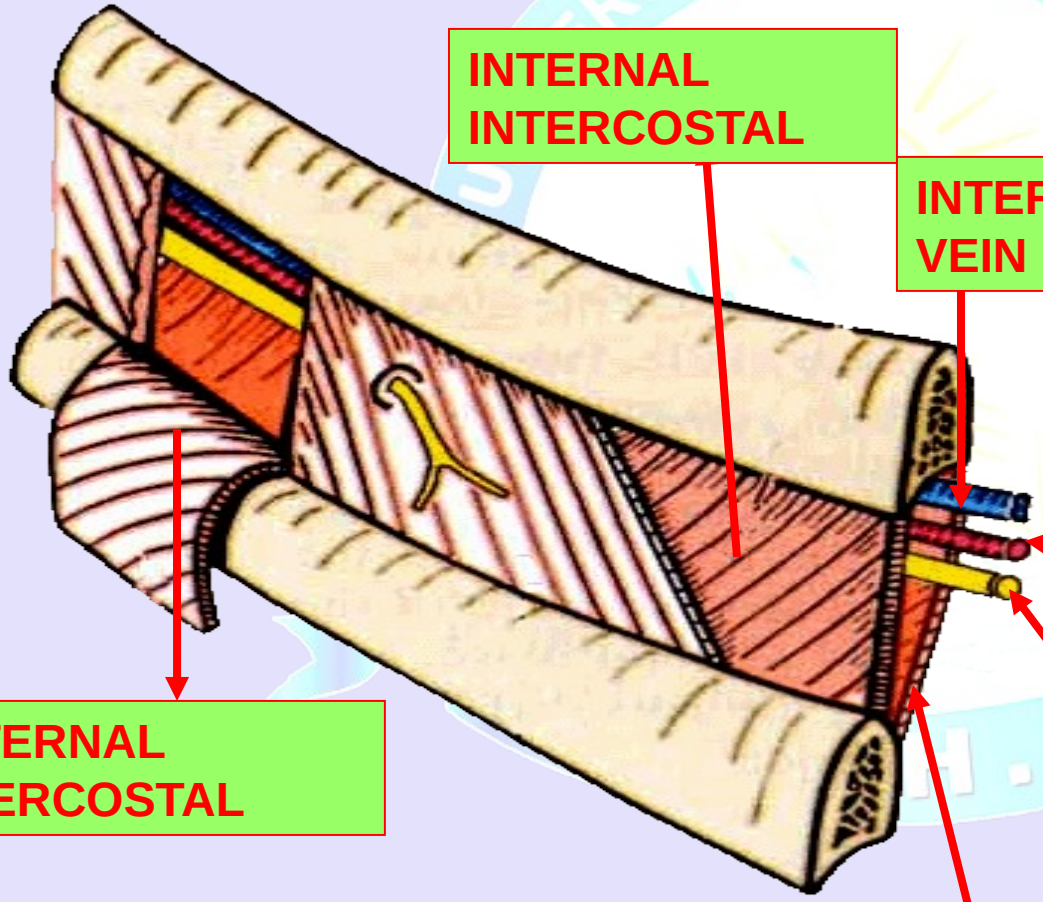
Thoracentesis performed to obtain sample for testing

INTERCOSTAL NERVE BLOCK

- In repair of laceration, fracture
- Intercostal nerve should be blocked before the lateral cutaneous arise
- Needle should be directed towards the rib near lower border, in subcostal groove
- COMPLICATION
- Pneumothorax
- hemorrhage



IDENTIFY??



INTERNAL INTERCOSTAL

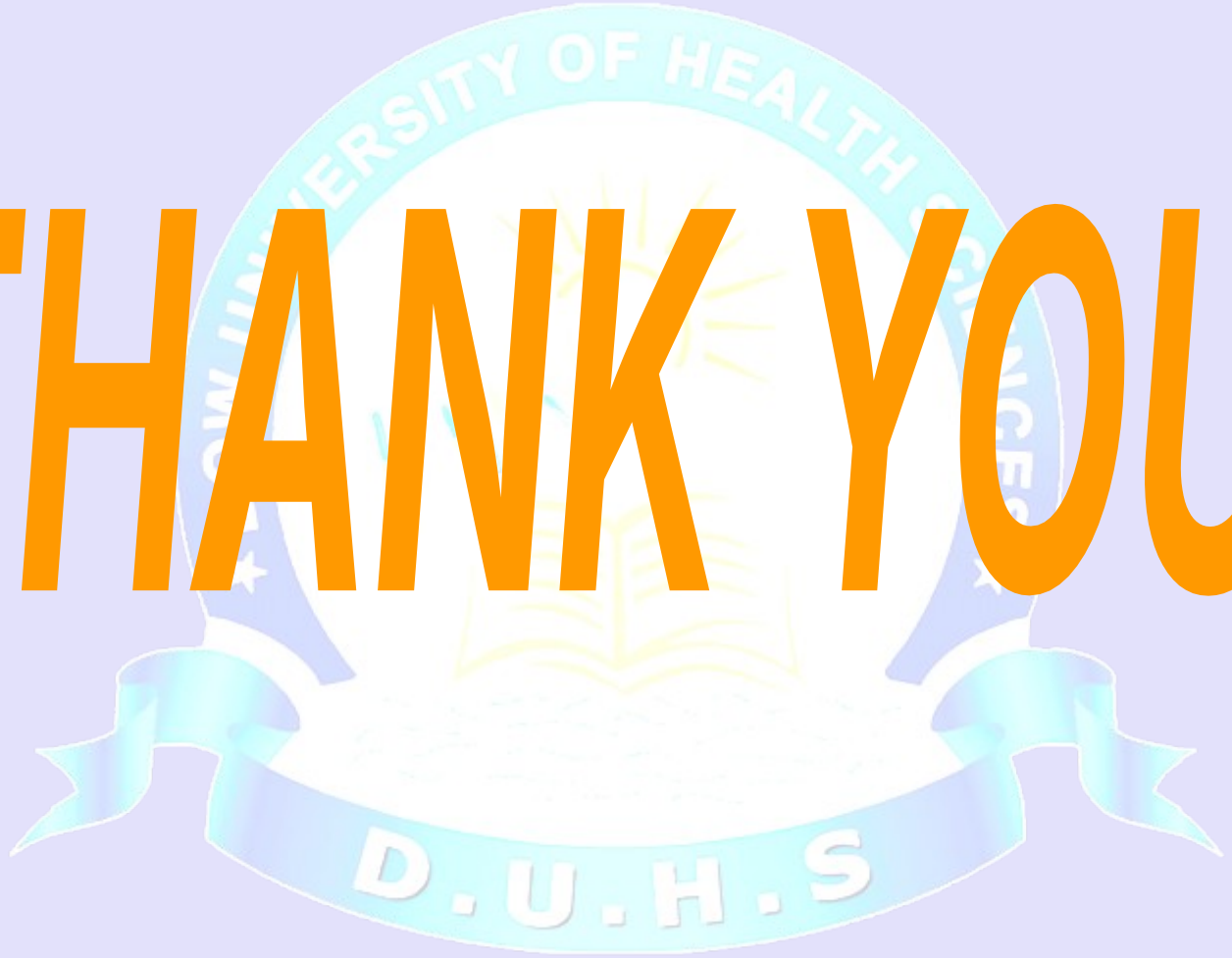
INTERCOSTAL VEIN

INTERCOSTAL ARTERY

EXTERNAL INTERCOSTAL

INTERCOSTAL VEIN

INNERMOST INTERCOSTAL



THANK YOU